FACULTY OF COMPUTING

SECP3843-01 SPECIAL TOPIC IN DATA ENGINEERING

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TITLE: TUTORIAL MS AZURE

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TABLE OF CONTENT

1. Project Overview

This project aims to address a significant business intelligence gap in centralized customer demographics and their impact on the product sales. The problem that is found in this project is that the organization possesses a large volume of customer data that is stored within an on-premise SQL database, however they lack consideration to analyze gender distribution and the influences of purchasing behaviour across many product categories. To overcome this issue, this project objective is to develop a comprehensive key performance indicator (KPI) dashboard which will provide a clear insight into total products sold and total sales revenue, specifically focusing on the gender split among customers. This project will create a strong automated data pipeline system which guarantees that organizations can access their most accurate and recent information needed for making strategic decisions.

2. Description on the solution

The implementation of hybrid cloud architecture required the development of an end-to-end data pipeline solution which operates on Azure cloud services. The process begins with a data ingestion process from a local SQL Server Express instance hosting the AdventureWorksLT2022 database. The system establishes a secure communication between its on-premises system and the cloud through a dedicated gateway in which it operates as a Self-hosted Integration Runtime (SHIR) solution. The system also uses Azure Key Vault to manage the sensitive credentials as the system requires database passwords and username to be stored securely while the Azure Data Factory pipeline retrieves these secrets dynamically.

The cloud-based data transfer process will start with secure data transfer to the cloud in which it enables processing and conversion of data through Azure Synapse Analytics into a format that is suitable for reporting. The deployment phase may face several difficulties which require the selected region for resource deployment to maintain compatibility with “Azure for Students” subscription. The solution also requires complete business requirements to be met through the process which feeds transformed data into the report while the entire pipeline operates on automatic daily schedule. The system will achieve high operational efficiency through this combined method as it applies cloud data engineering techniques in a professional manner.

3. Technology Use

This project utilizes a range of cloud-based services within Microsoft Azure which are Azure Data Factory, Azure Data Lake, Azure Databricks, Azure Synapse Analytics and also other technologies such as Power BI. Figure 1 shows how each technology is used in the data pipeline.

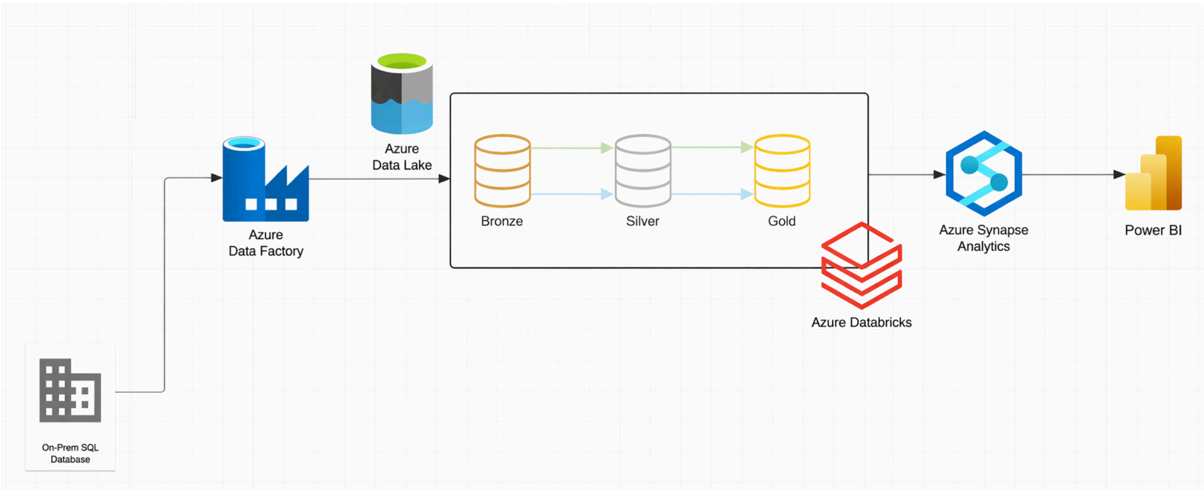

Figure 1 : *End-to-End Data Pipeline*

Azure Data Factory acts as the data integration tool which extracts the raw data from On-Prem SQL Database and loads and stores the data in a cloud storage. Azure Data Lake serves as the central repository for all the data in the system. The data lake supports the Medallion architecture by organizing data into Bronze, Silver, and Gold layers, thus enabling the storage of raw data, cleaned data, and curated data. For data cleansing, data processing and transformation, Azure Databricks is used in the data pipeline. Azure Synapse Analytics is used to perform complex queries on the processed data to ensure the data can be used for analysis and reporting. Lastly, Power BI is used to create interactive dashboards and visual reports by accessing the transformed data.

4. Data Pipeline Architecture

This project uses data pipeline architecture within Microsoft Azure ecosystem to visualize the systematic flow of information from the initial on-premises source through the three-tier Medallion Architecture which start from Bronze, Silver, and end at Gold which is ultimately crucial for business intelligence and reporting. Figure 2 shows a data pipeline architecture diagram that outlines the end-to-end data pipeline that is developed using Microsoft Azure ecosystem.

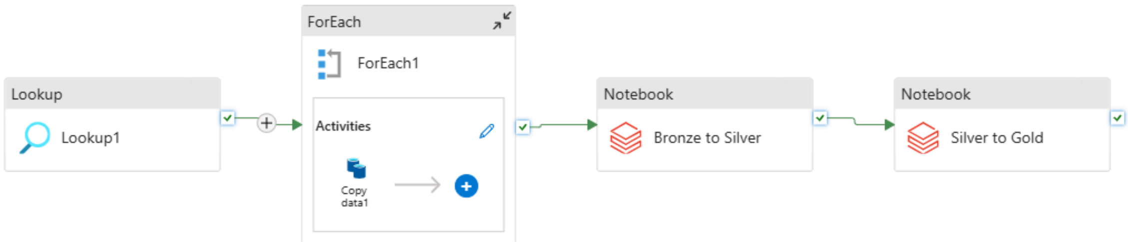

Figure 2: *Data Pipeline Architecture*

The data pipeline architecture follows the Medallion Architecture pattern which utilizes a multi-layered approach to transform the raw data into actionable business insights. The process starts with on-premises data ingestion where SQL Server Management Studio (SSMS) and SQL Server Configuration Manager are used to prepare the source of the database and ensure the necessary network protocols are enabled. To ensure the local environment and the cloud are linked together, a Self-Hosted Integration Runtime is configured through the Microsoft Integration Runtime. This setup ensures the Azure Data factory to securely access and extract data from on-premises SQL server, and land directly to the Bronze layer of Azure Data Lake Storage (ADLS) Gen2 account in raw and unfiltered format.

After the data is successfully ingested, Azure Databricks will start to manage the transformation process to move the data through Silver and Gold tiers. In the Silver layer, the raw data undergoes a cleaning and standardization process such as renaming the columns for consistency and correcting data types. In the Gold layer, it involves sophisticated data modeling where the cleaned data is transformed into a star scheme or aggregated tables for analytics. This enhanced data is then appeared to the end-users through implementation of Power BI where it provides a streamlined flow from initial local storage to high-level cloud visualization.

5. MS Power BI Dashboard

This project's motive is to develop a dashboard which provides an insight into total products sold and total sales revenue, focusing on the gender split among customers. Thus, an overall interactive dashboard with the requirements stated above have been created which is shown in Figure 2. Two cards featuring number of products and total sales revenue, two slicers featuring customer title and product name, and one donut chart displaying customer title are used in Power BI to create the interactive dashboard. The relationship between the tables were edited so that the cross-filter direction of the relationships between all the tables are both ways, enabling the donut chart and cards to change values when either of the slicers are used. Figure 3 shows the output after the interactive dashboard filtering the number of products, total sales revenue and customer title based on the selected product which is 'Brakes'.

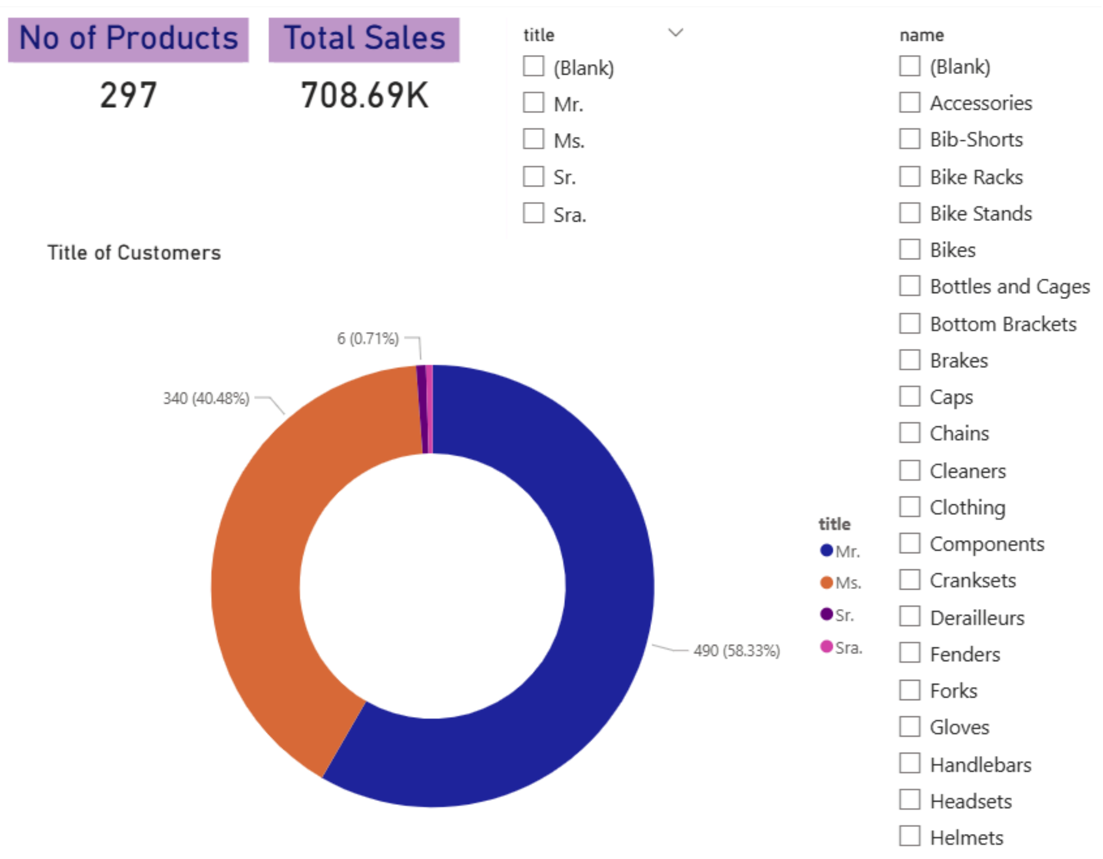

Figure 2 : *Overall Interactive Dashboard*

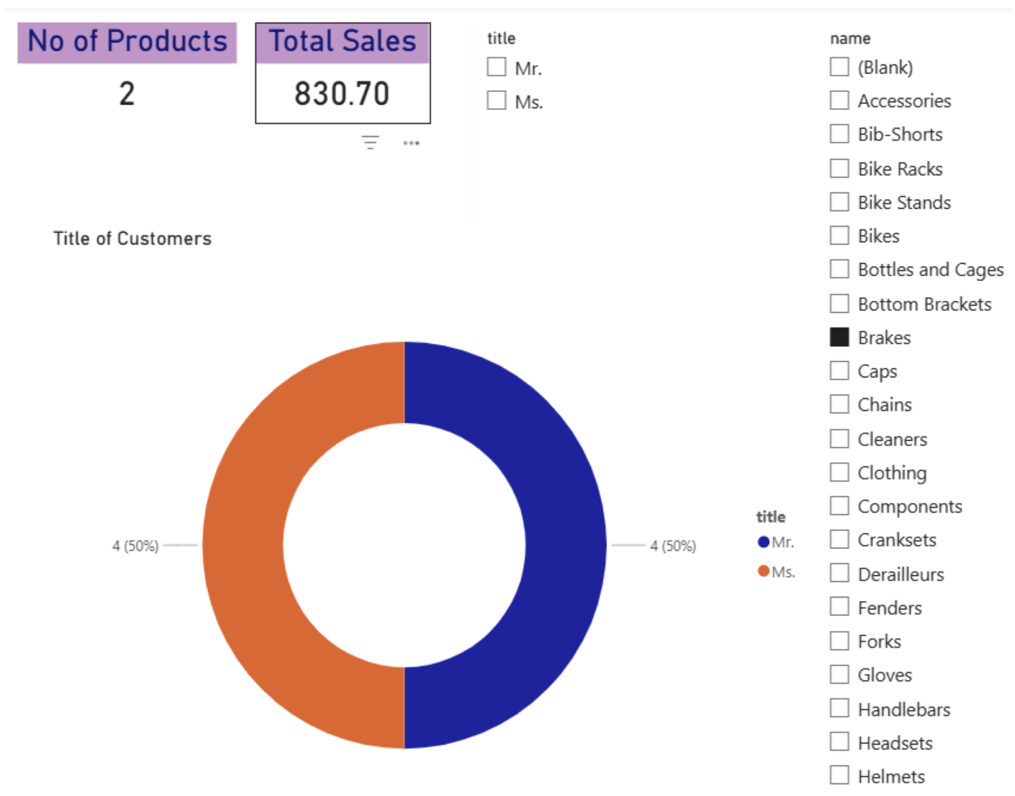

Figure 2 : *Interactive Dashboard After Filtering 'Brakes' Product*

6. Reflection

- **NABIL AFLAH BOO**

  Throughout this project, I’ve gained a great experience in building an end-to-end data pipeline by following the given Youtube tutorial, which allowed me to learn hands-on implementation using the theoretical data engineering concepts. Even though there were several technical issues of connecting Azure Data Factory at first, I had successfully managed the Azure Databricks to orchestrate data transformation across the Bronze, Silver, and Gold layers. Troubleshooting the Self-Hosted Integration Runtime and configuring SQL Server network protocols was the quite challenging part, however I managed to solve these connectivity issues with the help of Gemini. I also learned to build a dashboard using Power BI advanced features to transform the processed data into meaningful business insights.

- **HARINI**

  Through this project, I’ve learned to create an end-to-end data pipeline using Microsoft Azure and also create an interactive dashboard in Power BI. It was a little hard to follow along with the YouTube tutorial but I hope to master creating and managing the end-to-end data pipeline using Microsoft Azure with more practice in the future. Learning to use Power BI was relatively easy compared to Microsoft Azure features and I will try to learn more about the remaining features in Power BI on my own.